

Electrical Engineering Department
National Institute of Technology Kurukshetra

Subject: Power Electronics-1
Mid. Sem. - I

Time: 50 Minutes
Max. Marks: 15

Note: Attempt all questions.

Q1.	Demonstrate and describe the complete protection circuit of SCR in detail with neat and clean diagram.
Q2.	Describe the construction and working principle of the IGBT. Also, write the advantages of IGBT over other power electronics switches with proper reason. (a) Three SCRs are connected together to form a series string. The voltages across the thyristors are 320V, 300V and 280V respectively. What will be the value of equalising resistance to be used across each thyristor? (b) A single phase full converter charges a battery which offers constant voltage of 12V. A resistor R is inserted to limit the battery with charging current of 6A. Find the value of resistance R if supply voltage is 40V, 50Hz and conducting voltage drop across each SCR is 1V.
Q3.	